Market

Monopoly Market

1. Why is there no market supply curve under conditions of monopoly?

2. Define monopoly power. How can we measure the monopoly power by utilizing the Lerner Index? Briefly explain (Use relevant diagram to show the relationship between elasticity of demand and price markup).

3. What are the sources of monopoly power, and why some firms have more monopoly power than others? Explain.

4. Define price discrimination. Briefly explain the first, second, and third degree price discrimination (Use relevant diagrams).

Monopsony Market

5. How should a monopsonist decide how much of a product to buy? Will it buy more or less than a competitive buyer? Explain.

6. What is meant by the term "monopsony power"? Why might a firm have monopsony power even if it is not the only buyer in the market?

7. What are the some sources of monopsony power? What determines the amount of monopsony power an individual firm is likely to have?

8. Why is there a social cost of monopsony power? If the gains to buyers from monopsony power could be redistributed to sellers, would the social cost of monopsony power be eliminated? Explain Briefly.

Monopolistically Competitive and Oligopoly Market

9. What are the characteristics of a monopolistically competitive market? Briefly explain two sources of inefficiency in a monopolistically competitive industry (Use appropriate diagram).

10. What is Cartel? Explain the benefits and problems of forming a cartel.

11. Critically evaluate the 'Kinked Demand Curve Theory' of oligopoly market.

12. Let us consider a duopoly market where two firms produce a homogeneous product. The market demand is given by the inverse function,

$$\boldsymbol{P = 100 - 2(q_1 + q_2)}$$

Both the firm faces a constant MC of \$4. Now,
(i) Show the reaction function for both firms and the Cournot solution.
(ii) Compare the Cournot equilibrium with the perfect competitive equilibrium and when firms collude.
(iii) Graphically show the Cournot, competitive and collusive equilibrium.

13. Let us consider a duopoly market where two firms produce a homogenous product. The demand curve faced by the firms is $P = 100-0.5(q_1+q_2)$. The cost functions are $C_1=5q_1$, and $C_2=0.5q_2^2$.

(i) Find the reaction functions for both firms and find Cournot's solution.
(ii) Find the quasi competitive and collusion solutions and compare between them.

14 (a). Suppose there are two firms in a duopoly market. Firm-1's demand function is $Q_1=12-2P_1+P_2$ and Firm-2's demand function $Q_2=12-2P_2+P_1$. They face same fixed cost of \$20 and variable cost is zero for both firms. The firms play a Bertrand game.

(i) Derive and interpret the outcome of the game when the firm produce differentiated goods.
(ii) Instead of choosing the prices independently, what will happen if the two firms collude?

14 (b). Explain the meaning of a Nash equilibrium when firms are competing with respect to price. Why is the equilibrium stable? Why don't the firms raise prices to the level that maximizes joint profits?

Consumer Behavior

15. Illustrate and explain the followings for a normal good:
(i) Derivation of individual demand curve from price consumption curve (PCC).
(ii) Derivation of Engel curve from Income consumption curve (ICC).

16. Briefly explain the following assumptions which give the desired properties to the consumer's preference ordering:
(i) Completeness, (ii) Transitivity, (iii) Reflexivity, (iv) Non-satiation, (iv) Continuity, (vi) Strict convexity, and (vii) Differentiability

17. Assume that the utility function is $U = q_1q_2$ and the budget constraint is $M = p_1q_1+p_2q_2$.
(i) Construct the ordinary demand functions and deduce their properties.
(ii) Construct the Compensated demand functions.

18. Illustrate the income and substitution effects for normal, inferior, and giffen goods.

19. Suppose $U(x_1, x_2) = x_1x_2$ and $m = p_1x_1+p_2x_2$ are the utility function and budget constraint of a consumer respectively. Derive the Slutsky equation for a change in price of commodity x_1. Suppose $m = 400$, $p_1 = 4$, $p_2 = 10$ and equilibrium $x_1 = 50$, determine numerical values of substitution and income effect and interpret.

20. What does elasticity of substitution mean? Explain four simple production functions (Linear, Fixed proportions, Cobb-Douglas, and CES production function), each characterized by a different elasticity of substitution.

21. Briefly explain the properties of Profit function and Cost function.

22. Consider a consumer with the utility function $U = xy$, who faces a budget constraint of B and its given prices P_x and P_y.
(a) Set the choice problem and derive the consumer's Marshallian demand functions.
(b) Derive the indirect utility function for this problem.
(c) What is Roy's identity? Verify the Roy's identity for this problem.
(d) Set the choice problem and derive the consumer's Hicksian demand functions.
(e) Calculate the minimum-value function or expenditure function and verify Shephard's Lemma.